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A Census-Based Stereo Matching Algorithm with Multiple Sparse Windows

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Abstract—**Stereo matching is one of the most active research areas in intelligent vehicle technology. In order to apply the stereo matching to intelligent vehicles, it must generate high-accuracy three-dimensional information in real time. For real-time stereo matching, this paper proposes a sparse multi-window method which not only gives robustness to noise but also reduces hardware cost with high matching accuracy. This is achieved by reusing the result of overlapped operations between adjacent windows. The experimental results show that the proposed method can reduce the hardware complexity of stereo matching processors with higher accuracy compared with the conventional window method.**

Keywords—**Intelligent vehicle, stereo matching, census transform, sparse multi-window.**

I. INTRODUCTION

Stereo matching is a core technology of intelligent vehicles because stereo matching can obtain three-dimensional (3-D) information of the environment around a vehicle, such as distance or object information, by using a low cost stereo camera sensor [1-2]. The stereo matching technology applied to vehicles should guarantee safety. For the safety, stereo matching must present high-accuracy 3-D information and perform in real time [3].

Stereo matching is classified into two groups: local and global matching algorithms. The global matching algorithms tend to be more accurate than local matching algorithms, but the global matching algorithms are hard to implement for real-time performance because of its high computational overhead. Thus, the global matching algorithms are not suitable for real-time applications. On the contrary, the local matching algorithms are generally efficient and easy to implement [3-5]. The census transform is mostly used local matching algorithm because of relatively simple computation and robustness to brightness variations [6]. For highly accurate matching results of the census transform algorithm, a large window is required because the matching accuracy of census transform algorithm strongly depends on the length of the census vector. However, the length of the census vector is proportional to the amount of computation and hardware resources. To address this problem, several studies have developed methods that modify or simplify the original full census transform. Reference [7] presented a sparse census transform that simplify the original full census transform and showed optimized stereo matching results by software implementation. Reference [8] proposed mini census transform that uses fewer pixels for the census transform. Also, it presented hardware architecture based on ASW algorithm and showed the real-time performance. However, these methods use reduced census vectors for low hardware complexity, thus causing matching results to be corrupted.

This paper presents a sparse multi-window method which helps the improvement of matching accuracy and reduces hardware complexity without reducing the census vector for the census transform based stereo matching. The remainder of this paper is organized as follows. Section II explains the conventional sparse census transform method and proposes a sparse multi-window method. Experimental results are presented in Section III, and the conclusions are drawn in Section IV.

II. SPARSE MULTI-WINDOW BASED CENSUS TRANSFORM

A. Sparse Census Transform

The census transform introduced by Zabih and Woodfill is one of non-parametric local transforms [9]. The census transform is based on local intensity relationship between the center pixel and neighbor pixels within a certain census window. When $(2m+1) \times (2n+1)$ is selected as the census window size, the transform is defined as (1).

$$T(x,y) = \underset{i=-n}{\overset{n}{\otimes}} \underset{i=-m}{\overset{m}{\otimes}} \xi(I(x,y), I(x+i, y+j)) \tag{1}$$

where $I(x,y)$ is the intensity of pixel (x,y), and the operator $\otimes$ denotes a bit-wise catenation. The relationship function ξ is defined as (2).

$$\xi(p_1, p_2) = \begin{cases} 0 & if\ p_1 \le p_2 \\ 1 & if\ p_1 > p_2 \end{cases} \tag{2}$$

The hamming distance is calculated by hamming function which determines the number of differences between two transformed census vectors. Hamming distance is defined as

$$C_d(x,y) = Hamm(T_l(x,y), T_r(x-d,y)), \tag{3}$$

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where *Hamm* is hamming function, and d is disparity, subscript l and r mean left and right respectively. After the hamming distance is computed, the final disparity is calculated using the winner-takes-all (WTA) method, shown as

$$D_{Final}(x,y) = \arg\min_{d\in R} C_d(x,y), \tag{4}$$

where R is a set of all possible disparity values.

For reducing the computational complexity of census transform, [7] proposed a sparse census transform. It computed the hamming distance for dedicated pixels instead of all pixels within the census window as shown in Fig. 1.

Sparse census transform $T_s(x,y)$ and its hamming function $C_{s,d}(x,y)$ are defined as

$$T_s(x,y) = \bigotimes_{i=-n}^{n} \bigotimes_{i=-m}^{m} \xi(I(x,y), I(x+i, y+j)) \tag{5}$$

and

$$C_{s,d}(x,y) = Hamm(T_{s,l}(x,y), T_{s,r}(x-d,y)), \tag{6}$$

where n and m are the set of pixels in the sparse window.

B. Sparse Multi-Window Method

The sparse census transform is a useful method to reduce the hardware cost of the stereo matching processors. But it causes the drop of matching accuracy by using reduced census vector. So, this paper proposes the sparse multi-window method which not only prevents the loss of census vector, but also reduces the hardware cost.

The sparse multi-window method builds a sparse multi-window (SMW) around a center window (CW) by integrating adjacent sparse windows (SW). The CW consists of the center pixels of each SW. When SW and CW sizes are $(2k+1)\times(2l+1)$ and $a\times b$ respectively, the transform $T_{SMW}(x,y)$ and its hamming distance $C_{SMW,d}(x,y)$ are defined as (7) and (8).

$$\begin{aligned} T_{SMW}(x,y) &= \bigotimes_{u}\bigotimes_{v} T(x+u, y+v) \\ &= \bigotimes_{u}\bigotimes_{v}\bigotimes_{i=-k}^{k}\bigotimes_{j=-l}^{l} \xi(I(x+u, y+v), I(x+u+i\times a, y+v+j\times b)) \\ \text{where,} \quad u &= \begin{cases} [-b/2+1,\ b/2] & , \ b \text{ is even} \\ [-(b-1)/2,\ (b-1)/2], & b \text{ is odd} \end{cases} \\ v &= \begin{cases} [-a/2+1,\ a/2] & , \ a \text{ is even} \\ [-(a-1)/2,\ (a-1)/2], & a \text{ is odd} \end{cases} \end{aligned} \tag{7}$$

$$C_{SMW,d}(x,y) = Hamm(T_{SMW,l}(x,y), T_{SMW,r}(x-d,y)) \tag{8}$$

The transform $T_{SMW}(x,y)$ is a bit-wise catenation of every transform T corresponding with SWs in the SMW. The final cost $C_{SMW,d}(x,y)$ is calculated by the hamming function between $T_{SMW,l}(x,y)$ and $T_{SMW,r}(x,y)$.

Fig. 2 shows an example of the sparse multi-window method. In this example, a SMW of 6×6 size is made by 4 SWs of 3×3 size and the CW has 4 center pixels. Also, more large SMWs can be generated as shown in Fig. 3. Fig. 3(a) shows a 9×9 SMW consists of 9 SWs of 3×3 size, and Fig. 3(b) shows a 10×10 SMW consists of 4 SWs of 5×5 size.

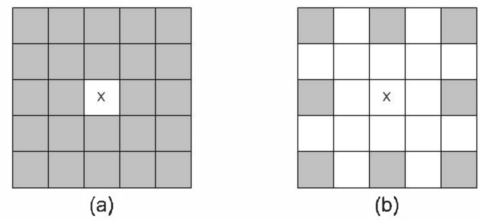

Figure 1. Window configurations: (a) The full census transform; (b) The sparse census transform.

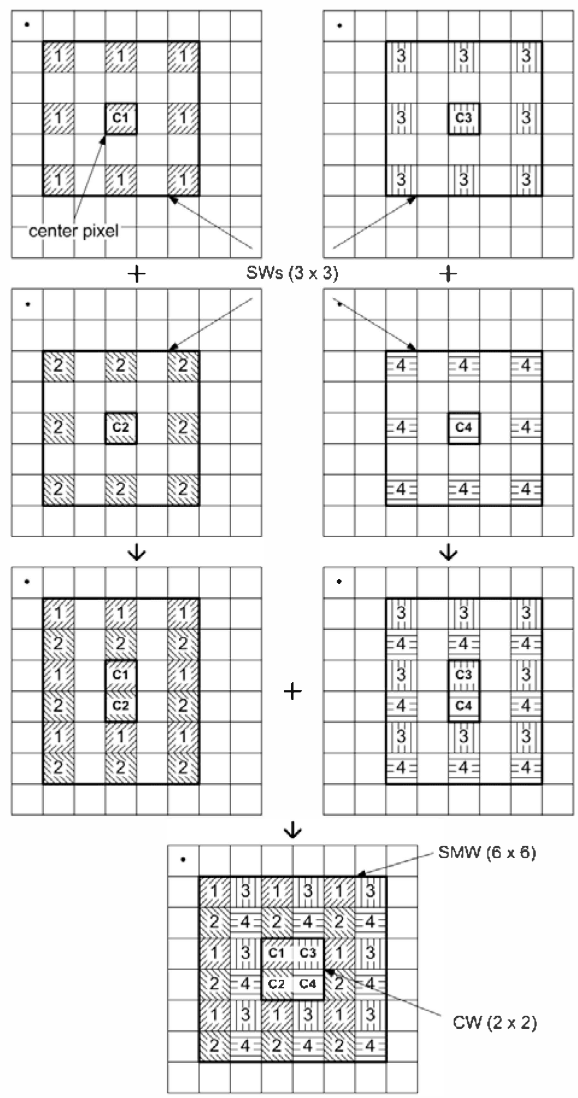

Figure 2. Illustration of the proposed sparse multi-window method.

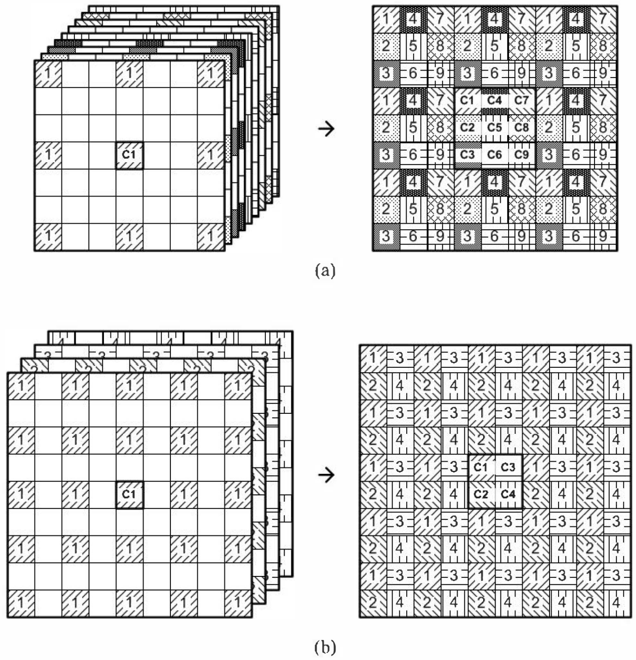

Figure 3. Examples of sparse multi-window methods: (a) 9×9 SMW consists of 9 SWs of 3×3 size; (b) 10×10 SMW consists of 4 SWs of 5×5 size.

The proposed sparse multi-window method has two advantages. First, it gives better robustness to noise than the conventional window method. In contrast with the usually used stereo test images, general camera images may have some noise that affects a bad influence on matching results. The proposed sparse multi-window method gives relatively accurate results compared with conventional window method, even if some of the center pixels are damaged by noise. This is mainly because, in the conventional window method, the center pixel damaged by noise can generate wrong hamming distance. On the other hand, the sparse multi-window method has multiple center pixels. Even if the corrupted hamming distances are generated by the damaged center pixels, the other center pixels generate the proper hamming distance. Therefore, the influence of noise can be covered.

Second, the proposed sparse multi-window method can be implemented with less hardware resources compared to the conventional window method. The proposed sparse multi-window method causes overlapped operations between adjacent SMWs and can reduce the hardware cost by reusing the results of the overlapped operations. In (1) and (7), the transform $T_{SMW}(x,y)$ is a bit-wise catenation of every transform result T in the SMW and each T is calculated independently. At this time, some T for $T_{SMW}(x,y)$ can be also used in $T_{SMW}(x+1,y)$. Fig. 4 shows the overlapped center pixels and SWs for T calculation between adjacent SMWs. According to Fig. 4, 1st SMW includes 4 center pixels (c1, c2, c3, and c4) and 2nd SMW also includes c3 and c4 center pixels, too. So, T corresponding with c3 and c4 in the 1st SMW can be reused in the 2nd SMW. Also, T corresponding with c5 and c6 in the 2nd SMW can be reused in the 3rd SMW in the same way.

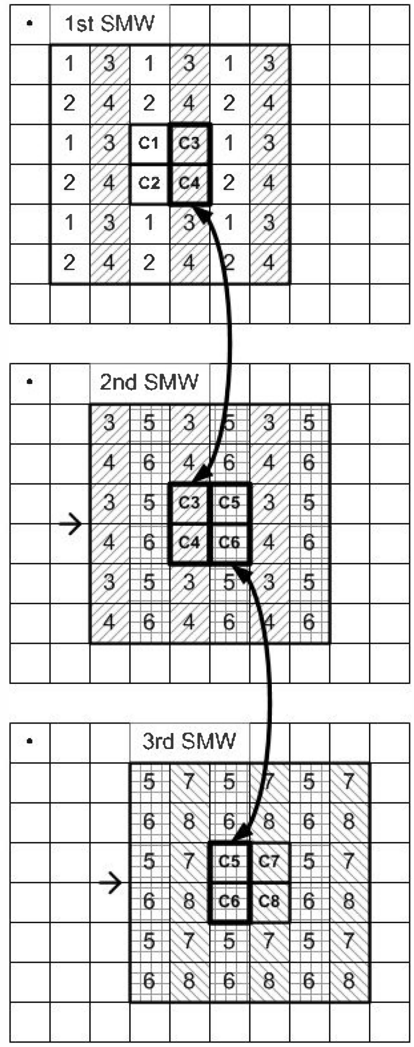

Figure 4. Overlapped center pixels and SWs between adjacent SMWs.

Using this method, the hardware cost for the transform and the hamming distance computation can be half without reducing the census vector. If the size of CW is larger, the more hardware resources can be reduced. For example, a 9×9 SMW consists of 9 SWs of 3×3 size as shown in Fig. 3(a) and the hardware cost is reduced to 1/3 compared with conventional window method.

III. Experimental Results

To evaluate the performance of the proposed method, we modeled the stereo matching processors based on the sparse multi-window method using C language and performed experiments with stereo image sets and their ground truth images. In addition, to show the hardware cost, stereo matching processors based on the proposed sparse multi-window method are implemented using HDL and verified on a FPGA-based platform. This section evaluates the effect of the sparse multi-window method on matching accuracy and hardware cost, and provides a discussion of the overall experimental results.

To investigate the matching accuracy of sparse multi-window configuration, this paper performed experiments using four types of Middlebury stereo datasets, shown in Fig. 5. The size of SWs and CWs directly relates to matching accuracy and hardware cost. Thus, we compared the matching accuracy and hardware cost using various sizes of SWs and CWs. The experiments set the size of CWs to 2×2, 3×3 or 5×5, and that of SWs to 7×7, 5×5 or 3×3. Fig. 6 and 7 show the comparison results of the matching accuracy ratio between the proposed methods and the conventional census transform with fixed size of SMW. For an objective comparison, the experiments set the size of SMWs as 14 or 15, and generate 3 types of SMW configurations for comparing with the conventional census window of 15×15 size. The results showed that the proposed sparse multi-window configurations outperform the general full window configuration. In addition, as the size of CW increases, the matching accuracy is improved. The major reason is that the multiple center pixels lead to blurring effect in results. So, lots of bad pixels are removed from results caused by census transform algorithm.

Fig. 8 shows the comparison results of the matching accuracy of the various types of SMWs. The results show the advantage of the multiple center pixels as discussed before. For overall SW sizes from 3×3 to 15×15, the best results are different according to the size of CWs. The best size of SWs is decreased as the size of CWs is increased. That means there is the best set of the CW and SW sizes according to the image. In contrast to common belief, the evaluations from the results exhibit the fact that very large SMWs lead to a decrease of the matching accuracy. This is mainly because the large SMW sizes broaden the object borders [3].

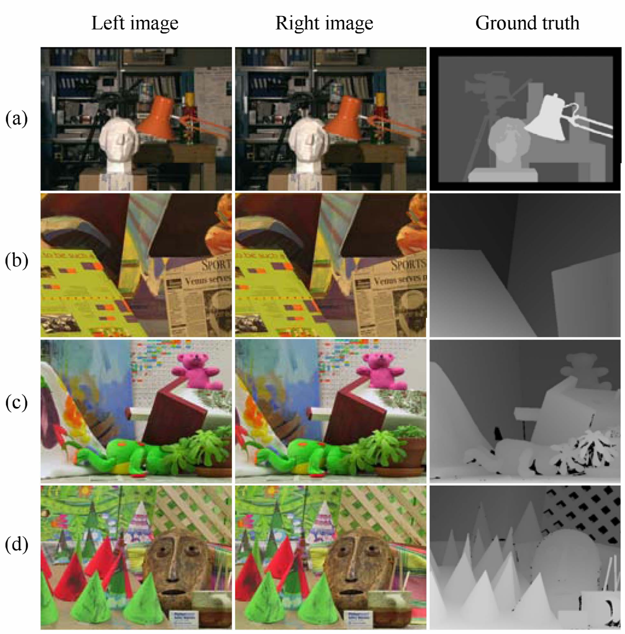

Figure 5. Middlebury stereo data sets used in the experiments: (a) Tsukuba, (b) Venus, (c) Teddy, and (d) Cones.

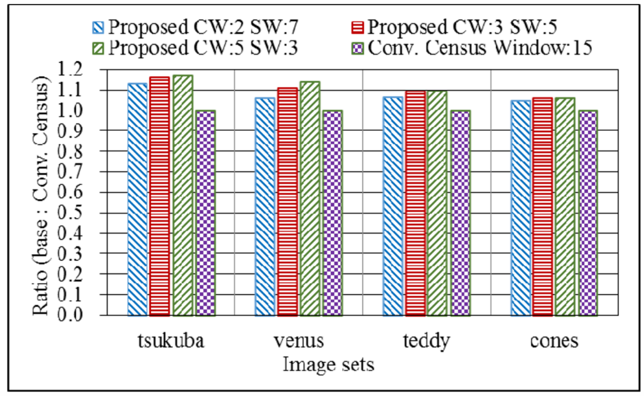

Figure 6. Comparison results of the matching accuracy ratio between proposed methods and conventional census transform.

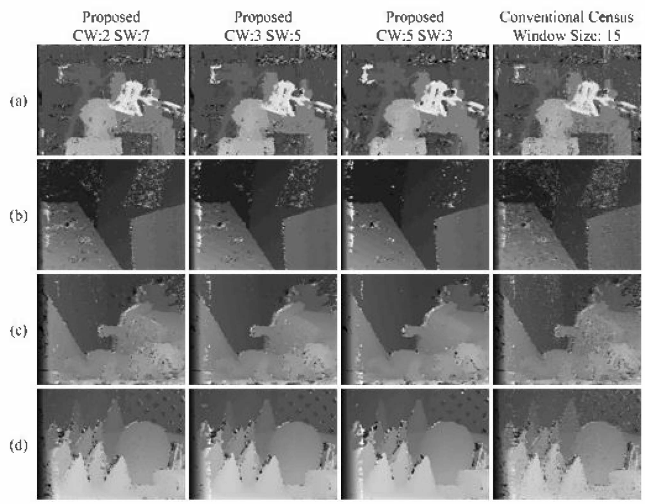

Figure 7. The result images of the proposed methods and conventional census transform.

Next, we investigated the matching accuracy of the proposed method with the noisy image sets. The Gaussian noise and the salt and pepper noise are inserted to the 4 Middlebury stereo datasets. Fig. 9 shows the comparison results of the matching accuracy of the various types of SMWs using the noisy images. The shape of experimental results is similar to that of Fig. 6 but the differences of accuracy rate in Fig. 9 are bigger than those of Fig. 6. The result means that the SMW gives robustness to noise.

Finally, in order to demonstrate the advantages of our method in terms of hardware cost, four types of census transform based stereo matching processors are designed using HDL and implemented on a Virtex-4 XC4VLX200-10 FPGA from Xilinx. Implemented stereo matching processors are as follows;

- Conventional census transform based stereo matching processor proposed by [10] (window size:15)
- Proposed census transform based stereo matching processor with SMW (CW:2, SW:7)

- Proposed census transform based stereo matching processor with SMW (CW:3, SW:5)
- Proposed census transform based stereo matching processor with SMW (CW:5, SW:3)

All of these processors are able to process up to 325 640×480 disparity maps per second running at 100 MHz clock frequency. Thus, the processing speed comparison of these processors is meaningless and not discussed in the experiments.

Table 1 summarizes the results of the implementation. The results showed that the proposed methods consumed more than 30% fewer hardware resources than the conventional census transform. In addition, as the size of CW increases, the hardware cost is decreased. This is because, as the size of CW increases, the overlapped operations which can be eliminated between adjacent SMWs are increased. When the best size of SMW is selected, the bigger size of CW is more suitable for hardware implementation.

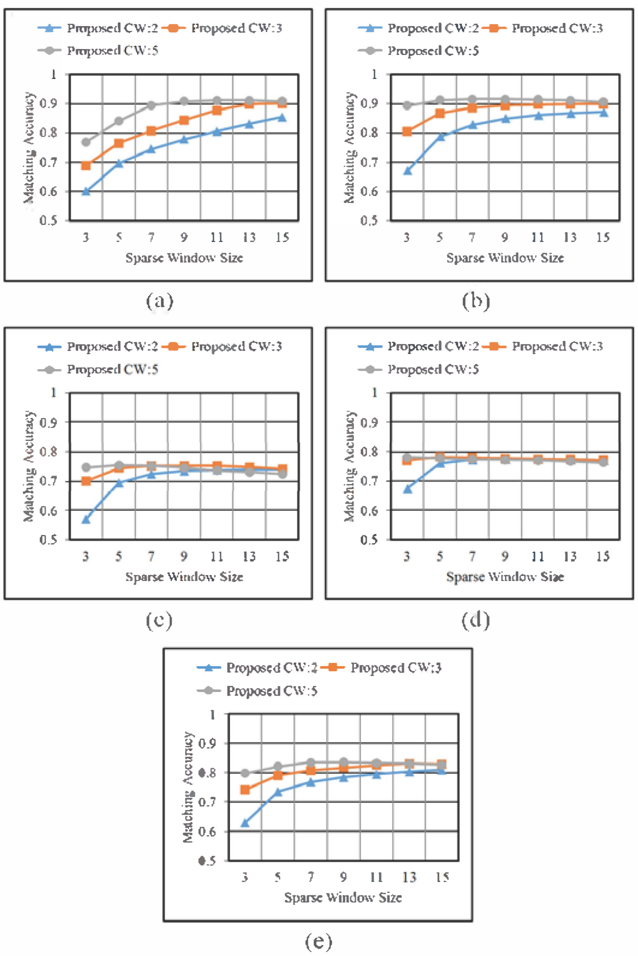

Figure 8. Comparison results of the matching accuracy of the various SW and CW configurations: (a) Tsukuba, (b) Venus, (c) Teddy, (d) Cones, and (e) Average

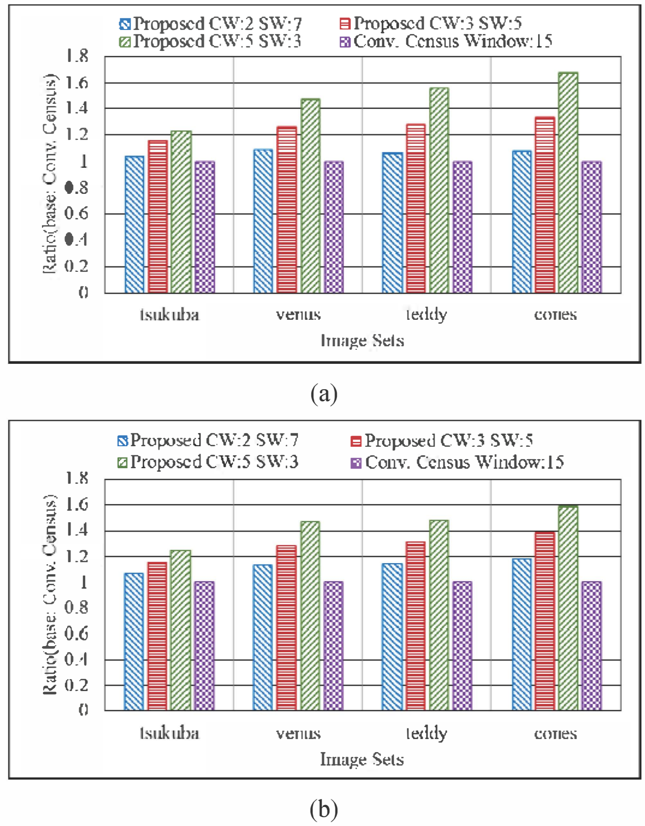

Figure 9. Comparison results of the matching accuracy ratio between proposed methods and conventional census transform when the noisy image was used: (a) Gaussian noise, (b) Salt and pepper noise.

TABLE I. HARDWARE COST SUMMARY

	Slice Flip Flop	*Ratio*	*LUTs*	*Ratio*	*Block RAM*	*Ratio*
Conv.Census	20,227	1	46,719	1	32	1
Proposed CW:2 SW:7	13,571	0.67	29,951	0.64	32	1
Proposed CW:3 SW:5	12,013	0.59	22,095	0.47	32	1
Proposed CW:5 SW:3	8,107	0.40	15,103	0.32	32	1

IV. CONCLUSION

Most important requirements in stereo matching applied to intelligent vehicles are high accuracy of resultant depth maps and real-time performance. So, this paper proposed the sparse multi-window method with census transform algorithm which not only increases the matching accuracy, but also reduces the hardware cost. The proposed sparse multi-window method causes overlapped operations between adjacent SMWs and can reduce the hardware cost by eliminating the overlapped operations. To evaluate the performance of the proposed method, this paper implemented the stereo matching processors based on the sparse multi-window method and conventional window method using C and HDL. The results show that the proposed method generated 10% more accurate results and consumed more than 30% fewer hardware resources than the conventional window method. In addition, the proposed method is robust to noise, and so suitable for real-time applications using the stereo matching.

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